

4.6 Vectors

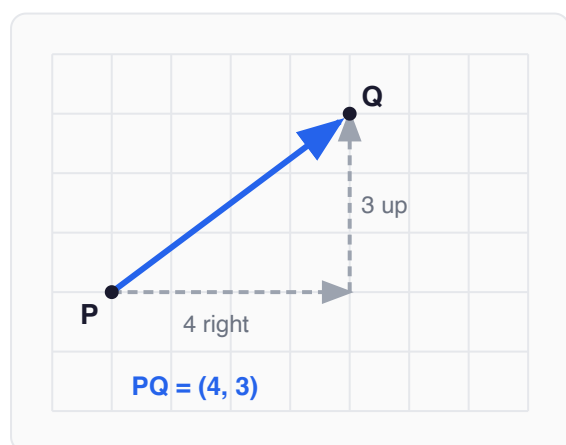
Pearson Edexcel GCSE Mathematics 1MA1 — Geometry and measures — spec refs G24, G25 — Papers 1 (non-calculator), 2 and 3 (calculator)

4.6.1 Vectors and Column Vectors (G25)

A **vector** describes a movement: it has a **magnitude** (how far) and a **direction** (which way).
A **scalar** is an ordinary number with size only, such as 5, -3 or $\frac{1}{4}$. Multiplying a vector by a scalar changes its length (and reverses it if the scalar is negative).

You will meet three ways of writing a vector on 1MA1 papers:

- **Column vector:** $\begin{pmatrix} 4 \\ -1 \end{pmatrix}$ — the top number is the horizontal movement (+ right, $-$ left) and the bottom number is the vertical movement (+ up, $-$ down).
- **Bold letter: $\mathbf{a}$.** When you write by hand, show it is a vector by **underlining**: $\underline{\mathbf{a}}$.
- **Two capital letters with an arrow:** $\overrightarrow{PQ}$ is the journey that starts at P and ends at Q .



Column vector from $P(x_1, y_1)$ to $Q(x_2, y_2)$ — “end minus start”:

$$\overrightarrow{PQ} = \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix}$$

Going the other way: $\overrightarrow{QP} = -\overrightarrow{PQ}$

Worked Example: A is the point $(-3, 2)$ and B is the point $(1, -4)$. Write down $\overrightarrow{AB}$ and $\overrightarrow{BA}$.

$$\overrightarrow{AB} = \begin{pmatrix} 1 - (-3) \\ -4 - 2 \end{pmatrix} = \begin{pmatrix} 4 \\ -6 \end{pmatrix} \quad \text{and} \quad \overrightarrow{BA} = \begin{pmatrix} -4 \\ 6 \end{pmatrix}$$

Equal Vectors and Negative Vectors

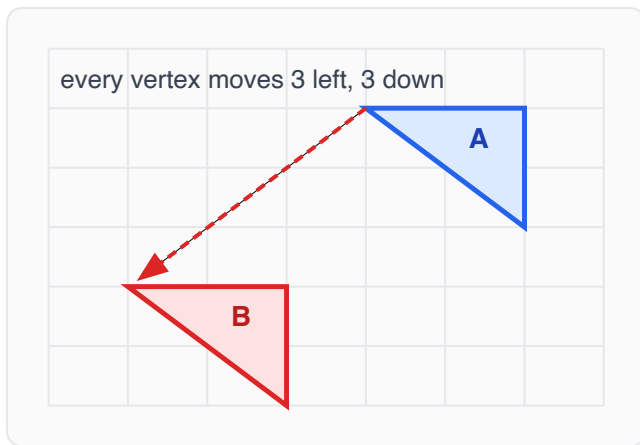
Vectors are **equal** when they have the same length *and* the same direction, wherever they are drawn. In a parallelogram $PQRS$, $\overrightarrow{PQ} = \overrightarrow{SR}$ because opposite sides are parallel and the same length.

$-\mathbf{a}$ is the same length as $\mathbf{a}$ but points the **opposite way**: if $\mathbf{a} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$ then $-\mathbf{a} = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$ — change the sign of both components.

4.6.2 Describing Translations with Vectors (G24)

A **translation** moves every point of a shape the same distance in the same direction. The image is **congruent** to the original and is the same way round. To describe a translation fully, give the word **translation** and its **column vector**.

The translation $\begin{pmatrix} a \\ b \end{pmatrix}$ sends the point (x, y) to $(x + a, y + b)$. To undo it, use $\begin{pmatrix} -a \\ -b \end{pmatrix}$.



Triangle A is mapped onto triangle B by the **translation** $\begin{pmatrix} -3 \\ -3 \end{pmatrix}$, and B is mapped back onto A by $\begin{pmatrix} 3 \\ 3 \end{pmatrix}$.

Worked Example: A triangle has vertices $(2, -1)$, $(4, -1)$ and $(2, 3)$. It is translated by $\begin{pmatrix} -5 \\ 3 \end{pmatrix}$. Write down the vertices of the image.

Subtract 5 from each x -coordinate and add 3 to each y -coordinate:

$$(2, -1) \rightarrow (-3, 2), \quad (4, -1) \rightarrow (-1, 2), \quad (2, 3) \rightarrow (-3, 6)$$

Worked Example: A shape is translated by $\begin{pmatrix} 4 \\ -1 \end{pmatrix}$ and then by $\begin{pmatrix} -6 \\ -3 \end{pmatrix}$. Write down the single translation with the same effect.

Add the two vectors: $\begin{pmatrix} 4 + (-6) \\ -1 + (-3) \end{pmatrix} = \begin{pmatrix} -2 \\ -4 \end{pmatrix}$

Exam Tip: Edexcel mark schemes for "Describe fully the single transformation" give one mark for **translation** and one for the correct **column vector**. Writing $(-3, -3)$ as a coordinate, or saying "moved", loses marks, and giving two transformations scores nothing. Count squares from a vertex to the *same* vertex on the image.

4.6.3 Vector Arithmetic and Vector Diagrams (G25)

Column Vectors

Add or subtract the matching components: $\begin{pmatrix} a \\ b \end{pmatrix} \pm \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a \pm c \\ b \pm d \end{pmatrix}$

Multiply by a scalar k : multiply both components, $k \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ka \\ kb \end{pmatrix}$

Worked Example: $\mathbf{a} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$. Work out $3\mathbf{a} - 2\mathbf{b}$ as a column vector.

$$3\mathbf{a} = \begin{pmatrix} 9 \\ -3 \end{pmatrix}, \quad 2\mathbf{b} = \begin{pmatrix} -4 \\ 8 \end{pmatrix}$$

$$3\mathbf{a} - 2\mathbf{b} = \begin{pmatrix} 9 - (-4) \\ -3 - 8 \end{pmatrix} = \begin{pmatrix} 13 \\ -11 \end{pmatrix}$$

Worked Example: $\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -2 \\ 6 \end{pmatrix}$. Find the vector $\mathbf{r}$ such that $\mathbf{p} + 3\mathbf{r} = \mathbf{q}$.

Treat it like a linear equation: $3\mathbf{r} = \mathbf{q} - \mathbf{p} = \begin{pmatrix} -6 \\ 9 \end{pmatrix}$, so $\mathbf{r} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$

Worked Example: Find m and n such that $m \begin{pmatrix} 1 \\ 2 \end{pmatrix} + n \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 10 \\ -1 \end{pmatrix}$.

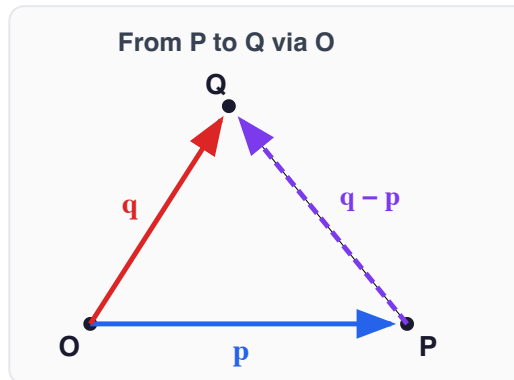
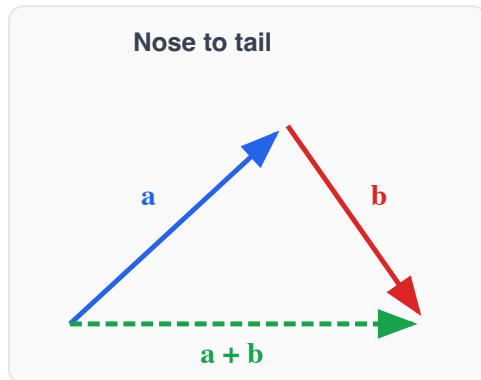
Top: $m + 3n = 10$. Bottom: $2m - n = -1$, so $n = 2m + 1$.

Substitute: $m + 3(2m + 1) = 10 \Rightarrow 7m = 7 \Rightarrow m = 1$, then $n = 3$.

Check: $\begin{pmatrix} 1 \\ 2 \end{pmatrix} + 3 \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 10 \\ -1 \end{pmatrix} \checkmark$

Vectors on Diagrams

To add vectors on a diagram, follow one and then the other: the **resultant** $\mathbf{a} + \mathbf{b}$ joins the start of the first to the end of the second. Subtracting $\mathbf{b}$ is the same as adding $-\mathbf{b}$, i.e. travelling *against* the arrow.



With $\overrightarrow{OP} = \mathbf{p}$ and $\overrightarrow{OQ} = \mathbf{q}$:

$$\overrightarrow{PQ} = \overrightarrow{PO} + \overrightarrow{OQ} = -\mathbf{p} + \mathbf{q} = \mathbf{q} - \mathbf{p}$$

This is “end minus start” again, now in letters instead of numbers.

Worked Example: $OABC$ is a parallelogram with $\overrightarrow{OA} = \mathbf{p}$ and $\overrightarrow{OC} = \mathbf{q}$. Find $\overrightarrow{OB}$, $\overrightarrow{CA}$ and $\overrightarrow{BC}$.

Opposite sides are equal vectors, so $\overrightarrow{AB} = \mathbf{q}$ and $\overrightarrow{CB} = \mathbf{p}$.

$$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = \mathbf{p} + \mathbf{q}$$

$$\overrightarrow{CA} = \overrightarrow{CO} + \overrightarrow{OA} = -\mathbf{q} + \mathbf{p} = \mathbf{p} - \mathbf{q}$$

$$\overrightarrow{BC} = -\overrightarrow{CB} = -\mathbf{p}$$

Worked Example: $ABCD$ is a parallelogram with $\overrightarrow{AB} = \mathbf{a}$ and $\overrightarrow{AD} = \mathbf{d}$. E is the midpoint of BC . Find $\overrightarrow{AE}$ and $\overrightarrow{DE}$.

$$\overrightarrow{BC} = \mathbf{d}, \text{ so } \overrightarrow{BE} = \frac{1}{2}\mathbf{d} \text{ and } \overrightarrow{AE} = \mathbf{a} + \frac{1}{2}\mathbf{d}.$$

$$\overrightarrow{DE} = \overrightarrow{DA} + \overrightarrow{AE} = -\mathbf{d} + \mathbf{a} + \frac{1}{2}\mathbf{d} = \mathbf{a} - \frac{1}{2}\mathbf{d}$$

Vector expressions obey the same rules as algebra: expand brackets and collect like terms.

Worked Example: Simplify $2(3\mathbf{a} - \mathbf{b}) - 4(\mathbf{a} - 2\mathbf{b})$.

$$= 6\mathbf{a} - 2\mathbf{b} - 4\mathbf{a} + 8\mathbf{b} = 2\mathbf{a} + 6\mathbf{b}$$

Exam Tip: Write the **route** before substituting, e.g. $\overrightarrow{DE} = \overrightarrow{DA} + \overrightarrow{AE}$. Moving **with** an arrow is $+$; moving **against** it is $-$. A correct route usually earns the first M1 even if the final simplification slips.

4.6.4 Parallel Vectors and Collinear Points (G25)

Higher

Two vectors are **parallel** when one is a **scalar multiple** of the other. $10\mathbf{a} - 4\mathbf{b} = 2(5\mathbf{a} - 2\mathbf{b})$, so it is parallel to $5\mathbf{a} - 2\mathbf{b}$ and twice as long. A negative multiple means parallel but pointing the **opposite way**; for column vectors, $\begin{pmatrix} -6 \\ 9 \end{pmatrix} = -3 \begin{pmatrix} 2 \\ -3 \end{pmatrix}$.

Points are **collinear** when they lie on one straight line. A complete collinearity proof has three parts:

1. Write two vectors that join the points, e.g. $\overrightarrow{AB}$ and $\overrightarrow{BC}$.
2. Show that one is a scalar multiple of the other, so they are parallel.
3. State that they have a **point in common**, so the points lie on the same line.

Worked Example: $\overrightarrow{AB} = 2\mathbf{a} - 5\mathbf{b}$ and $\overrightarrow{BC} = 6\mathbf{a} - 15\mathbf{b}$. Prove that A , B and C lie on a straight line.

$$\overrightarrow{BC} = 3(2\mathbf{a} - 5\mathbf{b}) = 3\overrightarrow{AB}, \text{ so } \overrightarrow{AB} \text{ and } \overrightarrow{BC} \text{ are parallel.}$$

Both pass through B , so A , B and C are collinear, with $AB : BC = 1 : 3$.

Worked Example: $4\mathbf{a} - 6\mathbf{b}$ is parallel to $k\mathbf{a} - 15\mathbf{b}$. Find k .

$-15\mathbf{b} = 2.5 \times (-6\mathbf{b})$, so the multiplier is 2.5 and $k = 2.5 \times 4 = 10$.

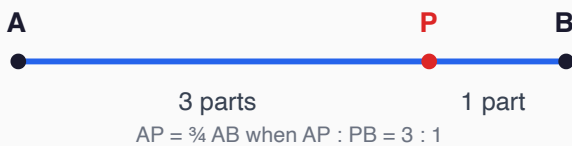
4.6.5 Midpoints and Points Dividing a Line in a Ratio (G25)

Higher

If P lies on AB with $AP : PB = m : n$, then P is $\frac{m}{m+n}$ of the way along:

$$\overrightarrow{AP} = \frac{m}{m+n} \overrightarrow{AB}$$

With $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$, the midpoint M of AB has $\overrightarrow{OM} = \mathbf{a} + \frac{1}{2}(\mathbf{b} - \mathbf{a}) = \frac{1}{2}(\mathbf{a} + \mathbf{b})$.



Worked Example: $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ and P lies on AB with $AP : PB = 3 : 1$. Find $\overrightarrow{OP}$ in its simplest form.

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a}, \text{ so } \overrightarrow{AP} = \frac{3}{4}(\mathbf{b} - \mathbf{a})$$

$$\overrightarrow{OP} = \mathbf{a} + \frac{3}{4}\mathbf{b} - \frac{3}{4}\mathbf{a} = \frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$$

Exam Tip: The ratio $m : n$ cuts the line into $m + n$ equal parts. Using $\frac{m}{n}$ is the classic error: for $AP : PB = 3 : 1$, P is $\frac{3}{4}$ of the way from A , not 3 times AB . A quick sense check — the coefficients in $\overrightarrow{OP}$ should add to 1 ($\frac{1}{4} + \frac{3}{4} = 1$).

4.6.6 Geometric Arguments and Proofs with Vectors (G25)

Higher

The Higher version of G25 adds “use vectors to construct geometric arguments and proofs”. Typical Edexcel questions ask you to prove that two lines are **parallel**, that three points are **collinear**, or to find a **ratio**. Work in the same order every time: express each vector you need in terms of the given vectors, simplify, then look for a scalar multiple and finish with a sentence.

Worked Example: OAB is a triangle with $\overrightarrow{OA} = 4\mathbf{a}$ and $\overrightarrow{OB} = 4\mathbf{b}$. P is on OA with $OP : PA = 1 : 3$ and Q is on AB with $AQ : QB = 3 : 1$. Prove that PQ is parallel to OB .

$$\overrightarrow{PA} = \frac{3}{4} \times 4\mathbf{a} = 3\mathbf{a}$$

$$\overrightarrow{AB} = 4\mathbf{b} - 4\mathbf{a}, \text{ so } \overrightarrow{AQ} = \frac{3}{4}(4\mathbf{b} - 4\mathbf{a}) = 3\mathbf{b} - 3\mathbf{a}$$

$$\overrightarrow{PQ} = \overrightarrow{PA} + \overrightarrow{AQ} = 3\mathbf{a} + 3\mathbf{b} - 3\mathbf{a} = 3\mathbf{b}$$

$$\overrightarrow{OB} = 4\mathbf{b}, \text{ so } \overrightarrow{PQ} = \frac{3}{4}\overrightarrow{OB}: \text{ a scalar multiple, so } PQ \text{ is parallel to } OB \text{ (and } PQ : OB = 3 : 4).$$

Worked Example: $OABC$ is a parallelogram with $\overrightarrow{OA} = 2\mathbf{a}$ and $\overrightarrow{OC} = 2\mathbf{c}$. M is the midpoint of CB and $\overrightarrow{OD} = 3\mathbf{a} + 6\mathbf{c}$. Prove that O , M and D are collinear.

$$\overrightarrow{CB} = \overrightarrow{OA} = 2\mathbf{a}, \text{ so } \overrightarrow{CM} = \mathbf{a} \text{ and } \overrightarrow{OM} = 2\mathbf{c} + \mathbf{a} = \mathbf{a} + 2\mathbf{c}$$

$$\overrightarrow{OD} = 3\mathbf{a} + 6\mathbf{c} = 3(\mathbf{a} + 2\mathbf{c}) = 3\overrightarrow{OM}$$

$\overrightarrow{OM}$ and $\overrightarrow{OD}$ are parallel and share the point O , so O , M and D are **collinear**, with $OM : MD = 1 : 2$.

Exam Tip: The final C1 (communication) mark needs a conclusion in words. “ $\overrightarrow{OD} = 3\overrightarrow{OM}$ ” on its own only shows the lines are parallel; add “and they share the point O , so O , M and D lie on a straight line.”

Summary

Skill	Key fact
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Translation (G24)	"Translation by $\begin{pmatrix} a \\ b \end{pmatrix}$ " — $(x, y) \rightarrow (x + a, y + b)$
Vector between points	End minus start; $\overrightarrow{QP} = -\overrightarrow{PQ}$
Column vector arithmetic	Add/subtract components; a scalar multiplies both components
Routes on a diagram	$\overrightarrow{PQ} = \mathbf{q} - \mathbf{p}$ when $\overrightarrow{OP} = \mathbf{p}$, $\overrightarrow{OQ} = \mathbf{q}$
Parallel Higher	One vector is a scalar multiple of the other
Collinear Higher	Parallel <i>and</i> a common point
Ratio $m : n$ Higher	$\overrightarrow{AP} = \frac{m}{m+n} \overrightarrow{AB}$